

way to decide which factors an investor should collect and determine how large his factor exposures should be.

A valid criticism is that I've outlined a great many sources of factor risk and a great many investable dynamic factor premiums. How do you choose in this cornucopia? How do you make sense of all these factor risk stories? The asset owner must take a stand by identifying what he believes and what defines his specific bad times. (An investor policy statement or a statement of investment beliefs are ideal vehicles to do this.) Then we compare the investor to the market along a select number of dimensions: time horizon, capacity to bear risk or weather losses, level and nature of liabilities, reliability of labor income, and so on. Start with the market portfolio—this represents the average investor. Judge how you are different from average along these characteristics, which determine your capacity to bear risk. Investors should choose just one or two dimensions to differentiate themselves from the market, translating their comparative advantages and disadvantages into risk factor exposures. The risk exposures are not necessarily positive—it might be better for us to purchase protection when volatility spikes and pay for volatility insurance, rather than receiving small volatility insurance premiums and getting whacked with heavy losses when volatility shoots sky high.

They're Your Assets

Agency problems are front and center in asset management. We don't want to be adding bad times through dysfunctional principal-agent relationships. This is an area where both industry and academia could do more in promoting the welfare of asset owners. In industry, investment vehicles too often give asset owners a raw deal. In many hedge fund and private equity contracts, for example, asset owners have little or no control over their funds, information is terrible, and agents have little incentive to act in the best interests of asset owners. Academics could do more by making recommendations for optimal contracts and deriving optimal control and governance mechanisms for asset management organizations.

Factor investing will help. Factor benchmarks can ameliorate agency issues. They do not penalize a manager for underperformance when the losses are due to poor factor performance. From the perspective of the asset owner, factors raise the bar for active management. There is talent out there, and we want to identify those with skill and reward them handsomely. However, that talent should be producing returns in excess of factor risk exposures that can be obtained generically at low cost elsewhere.

Currently, long-only *passive* factor benchmarks predominate in industry. The creation of low-cost factor portfolios should hasten the adoption of *dynamic* factor benchmarks tailor made for each investor. If we can get factors at low cost, we can remove from expensive active returns a component that doesn't really belong there.

Asset management is not really about the management of assets. It is all about factors.